MUHAMMAD ARRY NUGROHO

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Fresh Graduate in Geophysical Engineering at Bandung Institute of Technology (GPA: 3.38/4.00) with hands-on experience in field operations, geophysical data acquisition, equipment handling, data quality control, and technical problem-solving. Experienced in operating field equipment, supporting end-to-end field activities, and leading teams under field conditions through internship and academic projects. Strong interest in developing a career in oilfield field operations, with foundational knowledge of drilling and well control through IADC WellSharp and IWCF Level 1 training. Adaptable, safety-conscious, and eager to develop technical expertise through hands-on field experience.

Education Level

Institut Teknologi Bandung - Bandung — Aug 2022 - Oct 2026 (Expected)

Bachelor of Geophysical Engineering in Faculty of Mining and Petroleum Engineering, 3.38/4.00

- Undergraduate Thesis (2026): "Pore Pressure Prediction and Overpressure Zone Characterization in the Poseidon Field,Browse Basin, Australia using Eaton and Kan & Swan Methods through Well Log and Seismic Data Integration." Successfully created a 3D pore pressure distribution map with overpressure characterization of the Jamieson-Plover Formation at the Poseidon Field, Australia, using the Eaton and Kan & Swan methods.

Work Experiences

PT. Antam Tbk - Jakarta — Jun 2025 - Jul 2025

Exploration Geophysicist Intern

- Conducted field-based drone magnetic data acquisition using DJI Matrice 300/350 RTK integrated with a Geometrics MagArrow II UAV magnetometer.
- Operated and monitored field equipment while supporting acquisition planning, execution, and data quality control under field conditions.
- Performed end-to-end magnetic data processing, including diurnal and IGRF corrections, tie-line correction, gap adjustment, B-spline filtering, tie-line leveling, and microleveling.
- Produced Total Magnetic Intensity (TMI) maps and supported technical interpretation of subsurface magnetic anomalies.

Institut Teknologi Bandung - Bandung — Oct 2025 - Dec 2025

Practicum Assistant – Seismology

- Guided 31 students per semester through hands-on seismic analysis, including P- and S-wave picking, hypocenter and magnitude determination, and fault projection.
- Provided technical guidance in geophysical software and Python-based seismic data processing and analysis.
- Supported practical sessions by troubleshooting technical issues and ensuring students could execute procedures and interpret results correctly.

Institut Teknologi Bandung - Bandung — Oct 2024 - Jan 2025

Practicum Assistant – Geophysical Instrumentation 1

- Guided 31 students through hands-on geophysical instrumentation projects involving ESP32 microcontrollers and various sensors.
- Provided technical support in instrument design, assembly, testing, and troubleshooting to develop functional field-ready devices.
- Assisted students in applying measurement principles and operating instrumentation safely and effectively.

Organisational & Committee Experience

HIMA TG "TERRA" ITB - Bandung — Feb 2025 - Jan 2026

- Led the Interest and Talent Division, coordinating development programs and team members to deliver technical and professional development initiatives for 107 geophysical engineering students.
- Planned and executed programs while coordinating stakeholders, timelines, and team responsibilities.

Society of Exploration Geophysicists Student Chapter ITB - Bandung — May 2023 - Dec 2024

- Progressed from Staff of Research & Competition (2023) to Manager of Field Trip Division (2024), coordinating 3 industry exposure programs including SEG Field Camp, Company Visit, and Excursion for approximately 80 participants.

Character Development Training ITB - Bandung — Sep 2024 - Feb 2025

- Served as a trainer and mentor for 40 incoming ITB students, delivering training sessions and providing continuous guidance throughout the program.
- Coordinated with fellow trainers and participants to ensure effective communication, teamwork, and smooth execution of training activities.

Relevant Projects

- **Project - Gravity and Magnetic at Karangsambung** (2025): Served as a field operator for gravity and magnetic data acquisition in the Karangsambung area, covering the Jatibungkus region. Operated field instruments supported data acquisition and quality control, and performed end-to-end data processing. Contributed to geological interpretation of subsurface structures to identify potential intrusion zones.
- **Project - Aquifer Mapping** (2025): Led a 15-member field team in conducting resistivity surveys using a McOHM resistivity meter to map groundwater distribution in a rural village. Coordinated field activities and implemented real-time equipment safety and data quality protocols to maintain instrument integrity and reliable acquisition.
- **Project - Seismic Reflection Acquisition** (2024): Conducted seismic reflection surveys across 5 survey lines using a weight-drop seismic source, supporting field logistics, equipment handling, and data acquisition activities. Performed field data quality checks to support reliable and efficient acquisition.
- **Project - Geophysical Instrumentation** (2024): Designed and developed IoT- based geophysical instruments using ESP32, ADXL345, and INA219 sensors for seismic and resistivity measurements. Conducted instrument assembly, testing, and troubleshooting to develop functional field-ready devices.

Training and Certifications

- **IADC WellSharp Well Servicing – Introductory**
  Completed: Oct 2026 | Valid Until: Oct 2031 | Show Credential
  Key Competencies: Completion Operation • Well Barriers • Completion Equipment • Wireline • Coiled Tubing • Snubbing • Well Killing
- **IADC WellSharp Drilling Operations – Introductory**
  Completed: Sep 2026 | Valid Until: Sep 2031 | Show Credential
  Key Competencies: Well Control Fundamentals • Kick Prevention & Detection • Shut-In Procedures • Drilling Fluids • Pressure & Well Control Calculations • Well Killing Operations • Well Control Equipment • Abnormal Pressure Awareness
- **IWCF Level 1 Programme**
  Completed: Aug 2026 | Certificate No. L1P_13006729082026
  Key Competencies: Oil & Gas Fundamentals • Well Life Cycle • Drilling Operations • Well Control • Well Intervention & Workover • Pressure Control
- **KWS Web Services – Drilling Engineering**
  Completed: Sep 2026 | Valid Until: Sep 2027 | Show Credential
  Key Competencies: Drilling Fundamentals • Rig Equipment & Operations • Drill String • Drill Bit Fundamentals • Pore & Fracture Pressure • Directional Drilling • Well Control Fundamentals
- **English Language Proficiency – EF SET & TOEFL**
  EF SET: C1 Advanced | Score: 65/100 | Completed: Aug 2026 | Show Credential
  TOEFL ITP: Score 603 | Completed: Oct 2026 | Valid Until: Oct 2028 | Show Credential
  Key Competencies: Advanced English Proficiency • Reading Comprehension • Listening Comprehension • Written Communication • Spoken Communication
- **Data Analyst Bootcamp**
  Completed: Sep 2026 | Valid Until: Sep 2027 | Show Credential
  Key Competencies: Data Analysis • Data Cleaning & Transformation • Excel Data Analysis • SQL Querying • Python for Data Analysis • Power BI • Data Visualization • Data-Driven Decision Making
- **Python Bootcamp from Zero to Hero**
  Completed: Aug 2026 | Show Credential
  Key Competencies: Python Programming • Data Structures • File Handling • Automation • Excel & PDF Processing • Web Scraping • Image Processing • Computational Problem Solving
- **Anthropic AI Training – Claude 101 & AI Capabilities and Limitations**
  Completed: Sep 2026 | Show Credential | Show Credential 2
  Key Competencies**:** Generative AI • Prompt Engineering • AI-Assisted Workflows • AI Capabilities & Limitations • AI for Productivity

Skills

- **Technical Skills**: Field Data Acquisition, Equipment Operation, Data Quality Control, Equipment Safety, Seismic Acquisition & Interpretation, Well Log Analysis, Geomechanics, Pore Pressure Analysis, Drilling Operations & Well Control Fundamentals.
- **Software Skills**: Petrel, Hampson-Russell, Oasis Montaj, VISTA, MATLAB, Python, Qgis, Surfer, Res2DInv, IPI2Win, Microsoft Office, Power BI, and SQL.
- **Languages**: Indonesia (Fluent), English (Advanced).